Liquid Amount Sensing - Power Budget

Component	Voltage (V)	Current (mA)	Power (mW)	Source / Notes
PIC18F57Q43 Curiosity Nano	5	50	250	Conservative design assumption
NAU7802KGI	5	2.1	10.5	Datasheet operating current
1 kg Load Cell	5	15	75	Conservative design assumption
Status LED + R3	5	1.96	9.8	Calculated: (5 V - 2.65 V) / 1.2 kOhm
Tare/Zero Pull-up R4	5	0.5	2.5	Worst case with button pressed
R5/R6 Digital Pull-ups	5	2.13	10.65	Worst-case combined
Curiosity Nano External-Power Overhead	5	0.1	0.5	USB connected; board overhead
ESTIMATED TOTAL		71.79	358.95	
25% DESIGN MARGIN		17.95		
DESIGN REQUIREMENT		89.74		
FUSE LIMIT		500		
REMAINING CURRENT MARGIN		410.26		

LM7805 Thermal Budget

Condition	Input Voltage (V)	Output Voltage (V)	Design Current (A)	Regulator Dissipation (W)
Minimum Input	9	5	0.09	0.36
Maximum Input	12	5	0.09	0.63